DATA 201 · WEEK 8 HOMEWORK

Finding the Right Amount of Regularization

Investigate how regularization strength affects model performance, complexity, and interpretability.

Dataset: Hitters (same data and the same train/test split from the Week 8 lab). Build your work in a Google Colab notebook named **Week8_HW**.

Total: 14 points | **Instructor:** Rebin Muhammad, Montgomery College

Part 1 — Experiment with α (4 points)

Using the same training and test sets from the lab, fit three Lasso models with:

```
alpha_values = [0.1, 10, 100]
```

For each model, report:

- Training RMSE
- Test RMSE
- Number of nonzero coefficients

Put the three models in one comparison table.

Questions

1. As α increases, what happens to the number of retained predictors?
2. Does a smaller number of predictors always lead to a lower test RMSE?

Part 2 — Visualize and interpret (3 points)

Create a graph of α versus Test RMSE for the three models from Part 1.

Questions

1. Which α produces the lowest test RMSE among the three values?
2. Which α appears to cause the most underfitting? Explain using **bias**.
3. Why might a model with higher training RMSE produce lower test RMSE?

Part 3 — Model interpretation (3 points)

Using your results from Part 1:

1. Identify two predictors that Lasso sets to zero at $\alpha = 100$.
2. Does setting their coefficients to zero prove that these predictors have no relationship with salary? Explain.
3. If two models have very similar test RMSE but one uses fewer predictors, which would you prefer? Explain in terms of interpretability and predictive performance.

Part 4 — Choosing the right regression model (4 points)

For each scenario, choose **Ridge (L2)**, **Lasso (L1)**, **Ordinary Linear Regression**, or **Not Sure**. Explain your choice in 2–3 sentences. Consider the number of predictors, complexity, interpretability, and overfitting risk.

A. 80 patients, 200 medical measurements. Goal: identify a small number of measurements that may help predict disease severity.

B. 5,000 employees, 40 predictors. Many predictors are strongly correlated, and most are believed to contain useful information. Keeping all predictors is fine.

C. 10,000 observations, only 3 predictors. The relationship looks roughly linear. The researcher wants a simple, easy-to-interpret model.

D. 500 observations, 25 predictors. It is unclear how many predictors matter or how strongly they are correlated. The company wants the most accurate predictions possible.

Reminder: there is not always enough information to know which model will win. Say when cross-validation would be needed.

Submission

1. Create a Google Colab notebook named **Week8_HW**.
2. Include your Python code, results table, graph, and written answers.
3. Submit the completed notebook to **Microsoft Teams** for Week 8 Homework.

Tip: reuse the preprocessing and split from the Week 8 lab so your homework results stay comparable to the lab.